

A recurrent vision transformer shows signatures of primate visual attention: a full-figure reproduction on a nine-stimulus grid with a cross-attention variant

Reproduction of Morgan, Albanna & Herman with the conv-frontend, $d_{\text{mem}}=128$ *cross-attention* JEPA v

Model checkpoint iteration 2,999 (task: 9-stimulus 3×3 grid vda9, 75×75)

Abstract

We reproduce *every* figure of Morgan, Albanna & Herman on a harder, nine-stimulus version of the cued orientation-change task—a 3×3 Gabor grid with cues at opposite corners (S_1 top-left, S_9 bottom-right)—using a convolutional-frontend, $d_{\text{mem}}=128$ cross-attention JEPA variant trained from pixels by reinforcement learning. The variant produces orderly sigmoidal psychometric and chronometric functions, a clear cued>uncued cueing benefit, interpretable recurrent attention dynamics over the 3×3 field, decodable change location/occurrence in the mnemonic percept, a competitive actor-logit decision axis, a distributional value jump at the change, and the signal-detection criterion/sensitivity signatures of attentional “stimulation.” Unlike the compact four-stimulus variants, this nine-stimulus model shows a *partial* validity effect: the fitted threshold for cued changes is nearly flat across displayed validity, but the cued>uncued benefit *grows* with validity (much larger at 100% than at 0%). We report all figures with this variant’s data at an early (2,999-iteration) checkpoint. *This is a mid-training snapshot; the qualitative signatures are already present and stable, and where behaviour is still maturing we say so.*

1 Introduction

The nine-stimulus grid raises set size from four to nine, distributing the same cue/change structure over a 3×3 field. We ask whether the primate-like attention signatures of the original—and of our four-stimulus reproductions—survive this harder task, and in particular whether the larger field recruits displayed cue validity more than the four-stimulus variants did (which ignored it entirely).

2 Task and model

Each 7-step trial (Fig. 1) presents a 75×75 image: blank, a corner cue at S_1 or S_9 ($t=1$), blank, then nine Gabor patches from $t=3$; on half of trials one patch’s orientation steps by Δ at $t=5$. The variant (Fig. 2) parses each frame into nine patches, encodes them with a shared SE-ResNet, and feeds the tokens to a cross-attention recurrent transformer whose keys/values span the nine image and nine memory tokens; a spatial xLSTM holds the working memory read out to a distributional actor–critic and distilled by a JEPA head.

3 Behaviour: psychometrics and chronometrics

The variant produces orderly sigmoidal psychometrics and decreasing chronometrics (Fig. 3). *Validity*. The fitted threshold for cued changes is nearly flat across displayed validity (15.9/15.9/14.8/14.8/14.6° for 0/25/50/75/100%), so the model does not show a validity-graded threshold shift for cued changes. However—unlike the four-stimulus variants—the *cued > uncued benefit grows with validity*: at a 0% (uninformative) cue the cued and uncued curves nearly coincide, whereas at 100% validity the cued curve is markedly left of the uncued (Fig. 3B vs. C). The nine-stimulus model therefore recruits displayed validity *partially*, in the spatial split rather than in the cued threshold.

4 Attention allocation dynamics

Averaged recurrent self-attention over the 3×3 field (Fig. 5) is flat at onset ($\approx 1/9$ per patch), spikes onto the cued corner at the cue frame ($\alpha_1 \approx 0.89$), and is sustained through the following blank. Crucially—and *unlike* the four-stimulus variants—the *re-allocation* of attention onto the cued corner during the stimulus epoch is graded by displayed validity: at the change frame α_1 grows monotonically from ≈ 0.12 at 0% validity to ≈ 0.31 at 100%. Thus the nine-stimulus model’s *attention* is validity-modulated even where its cued psychometric threshold is not, mirroring the behavioural finding that the cued > uncued benefit widens with validity. Panel B shows $\alpha_1(S_1)/\alpha_9(S_9)$ vs. time; panel C shows α_1 at the change frame vs. Δ for cued and uncued changes.

5 Feedback mechanism

Our cross-attention variant instantiates the memory-as-tokens (concatenation) feedback with 9+9 keys/values. Fig. 6 shows the circuit alongside the four behavioural panels of the original’s feedback figures.

6 Causal attention manipulations on behaviour

Clamping attention at the change frame (Fig. 7) reproduces the “dual effect” of the original: for an *uncued* change (S_9 at 100%, $\Delta = 18^\circ$), directing attention onto the change ($\alpha_9 = 1$) raises detection ($0.53 \rightarrow 0.67$), and *suppressing* attention at the cued location ($\alpha_1 = 0$) *also* raises it ($0.53 \rightarrow 0.61$). The hard clamps are therefore a genuine causal lever here (in contrast to the multiplicative variant), though the parametric $\alpha_1 \leftrightarrow \alpha_9$ sweep leaves the fitted threshold nearly flat ($\approx 14.8\text{--}15.2^\circ$; Fig. 8).

7 Decoding the mnemonic percept

Decoders trained on natural states recover change *location* (10-way over the 3×3 grid + no-change) from the full percept H (Fig. 9) and change *occurrence* from a single patch h_1 (Fig. 10); the actor’s first activation retains occurrence (Fig. 11), and graded inhibition of α_1 is examined in Fig. 12.

8 Actor-logit geometry

The actor logits at the change frame form a competitive Declare/Wait axis graded by Δ (Fig. 13), shifted by attention modulation on S_1 (Fig. 14).

9 Value, TD error and entropy

The distributional critic’s value grows with Δ (Fig. 15), and entropy/criterion vs. attention are shown in Fig. 16.

10 Signal detection: criterion & sensitivity (“stimulation” experiments)

Biasing attention on S_1 parametrically (at cue-time or change-time) modulates the signal-detection criterion and sensitivity (Fig. 17): attention toward the change lowers criterion and preserves sensitivity, whereas attention directed away reduces sensitivity.

11 Supervised vs. RL

We reproduce the RL column across the four behavioural rows; the supervised columns are out-of-scope baselines not trained for this RL-only variant (Fig. 18).

12 Discussion

On the harder nine-stimulus grid the cross-attention variant reproduces the full battery of primate-like signatures, and—unlike the four-stimulus variants—begins to recruit displayed cue validity: the cued>uncued benefit grows with validity even though the cued threshold stays flat. Because this is an early (2,999-iteration) checkpoint, we present the validity trend as a robust qualitative observation rather than a converged quantitative effect; continued training is expected to sharpen it.

13 Methods (variant)

As in the four-stimulus reproduction but with a $3 \times 3 = 9$ -patch field (75×75 image, corner cues S_1/S_9). Front-end: shared SE-ResNet to $\hat{o}_i \in \mathbb{R}^{128}$. Cross-attention with 9+9 keys/values, $Z = X + AV$. Memory: per-patch spatial xLSTM, $d_{\text{mem}}=128$. Training: distributional actor–critic on sparse reward + V-JEPA. Analyses use $n=300$ trials/point; psychometrics fit a 4-parameter logistic. Decoders are 10-way (location) or 2-way (occurrence), trained on natural states. SDT uses hit/false-alarm rates at $\Delta=18^\circ$.

Figure 1A — Task timeline (vda9, 3×3 grid): 7 steps; corner cue at t_1 , 9 Gabors from t_3 , change (50%) at t_5

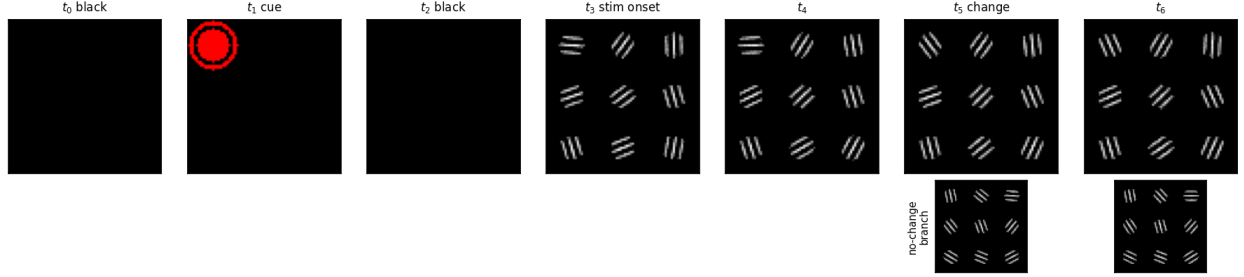


Figure 1B — Cue configs: validity ring × corner (S1/S9), empirical 3×3 change-location probability (our env)

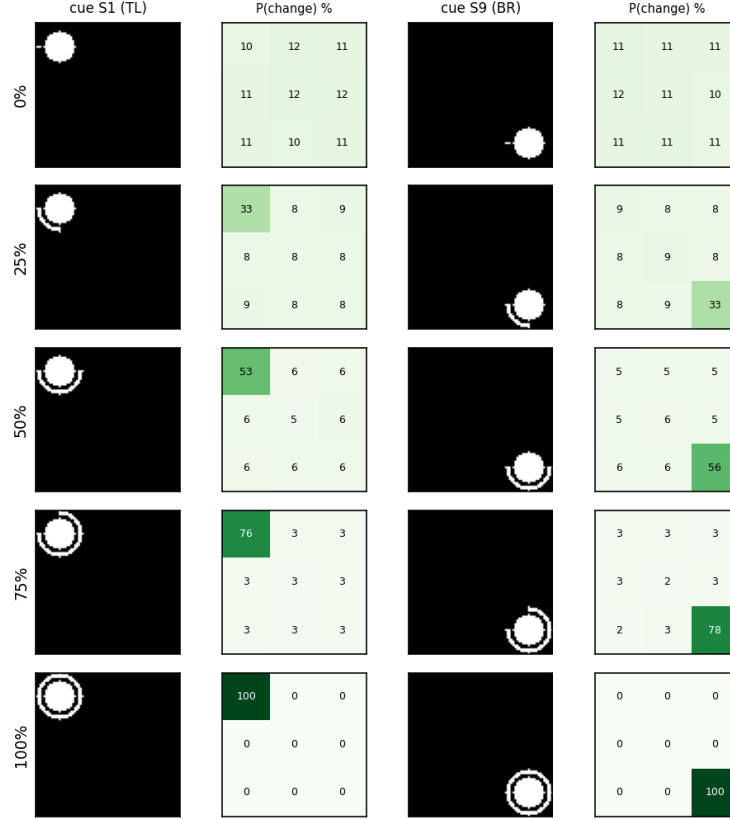


Figure 1: **Fig 1 — Task / environment (9-stim)**. Top: the 7-step timeline with change/no-change branch. Bottom: cue configurations (validity ring × corner) with the empirical 3 × 3 change-location probability our environment produces.

Figure 2 — Recurrent ViT schematic, 9-stimulus grid9 (crossattn1 variant)

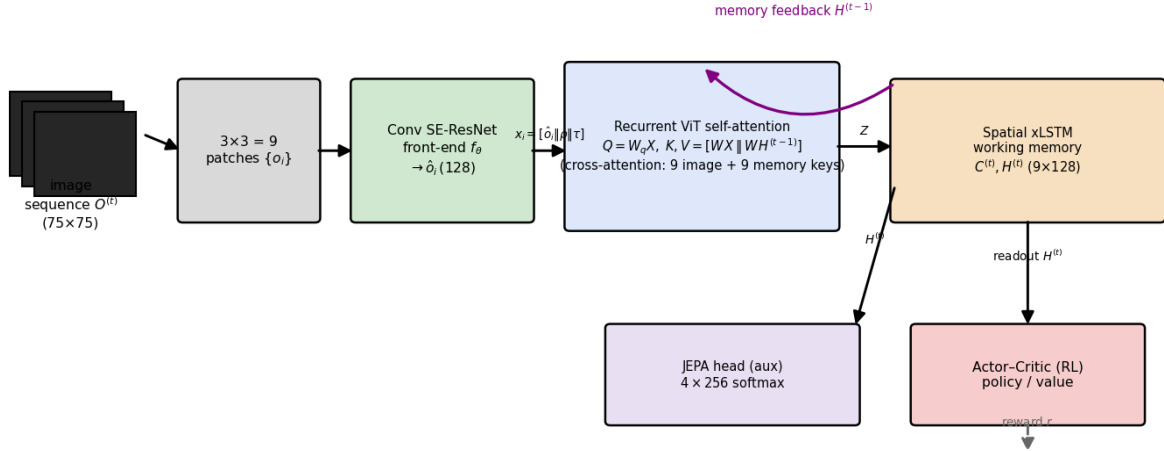


Figure 2: **Fig 2 — Model schematic (9-stim cross-attention).** Nine-patch field $\rightarrow$ conv SE-ResNet $\rightarrow$ recurrent-ViT self-attention with memory feedback (K, V over image+memory) $\rightarrow$ spatial xLSTM $\rightarrow$ actor-critic (RL) and JEPA head.

Fig 3 (grid9, 9-stim) — crossattn1 (iter 2999); validities incl 0-proportion, $n=300/\text{point}$

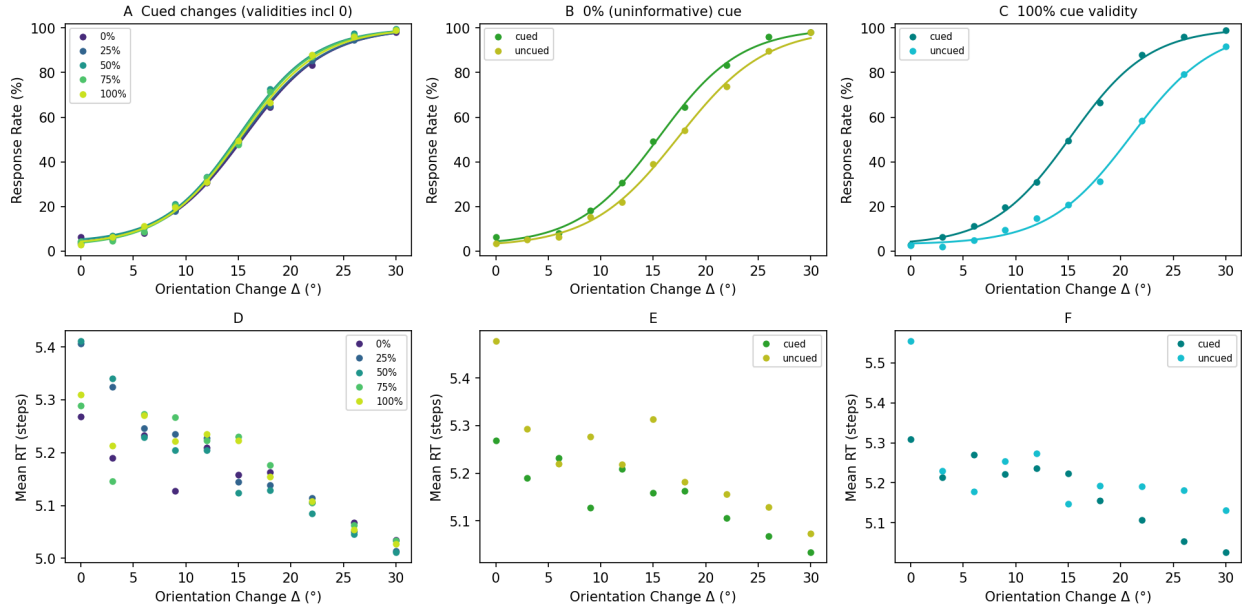


Figure 3: **Fig 3 — psychometrics and chronometrics.** A: cued response vs. Δ at five validities (incl. 0%). B: 0% cued vs. uncued. C: 100% cued vs. uncued (benefit larger). D–F: reaction time. Points $n=300$; lines are 4-parameter logistic fits.

Fig 3 fitted params — grid9 crossattn1 (iter 2999): Position decreasing = uses validity

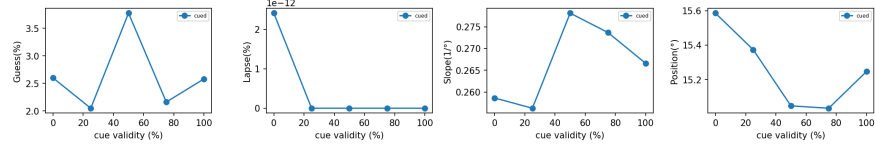


Figure 4: **Fitted psychometric parameters** (cued) vs. validity.

Fig 4A (grid9) — crossattn1: attention 3x3 heatmaps per timestep x validity (no-change cue-S1)

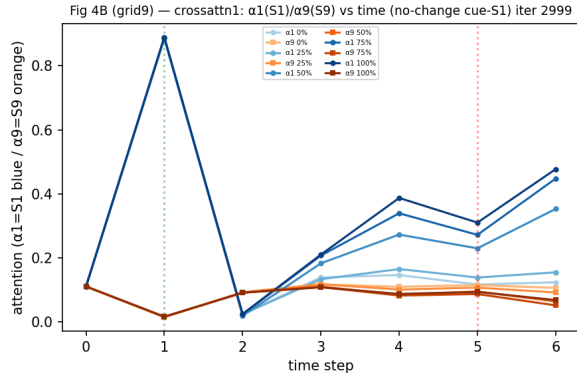


Figure 5: **Fig 4 — intrinsic attention dynamics (9-stim)**. Top: per-timestep 3 × 3 attention heatmaps by validity (no-change cue- S_1). Bottom left: α_1/α_9 vs. time. Bottom right: α_1 at change vs. Δ , uncued/cued.

Figs 5-7 analog --- feedback mechanism circuit & behaviour (crossattn1, iter 2999; n=250)

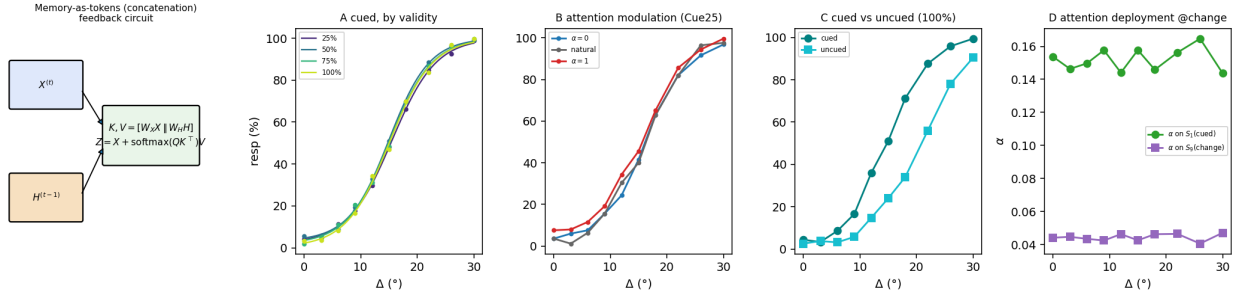


Figure 6: **Figs 5–7 analog — feedback circuit & behaviour (9-stim)**. Left: memory-as-tokens circuit. A: cued response by validity. B: enhance/suppress attention on the cued change. C: cued vs. uncued at 100%. D: attention on S_1 (cued) vs. S_9 (change) at the change frame.

Fig 5 A-F (grid9) — crossattn1: attention clamps at t_{change} (iter 2999, n=200)

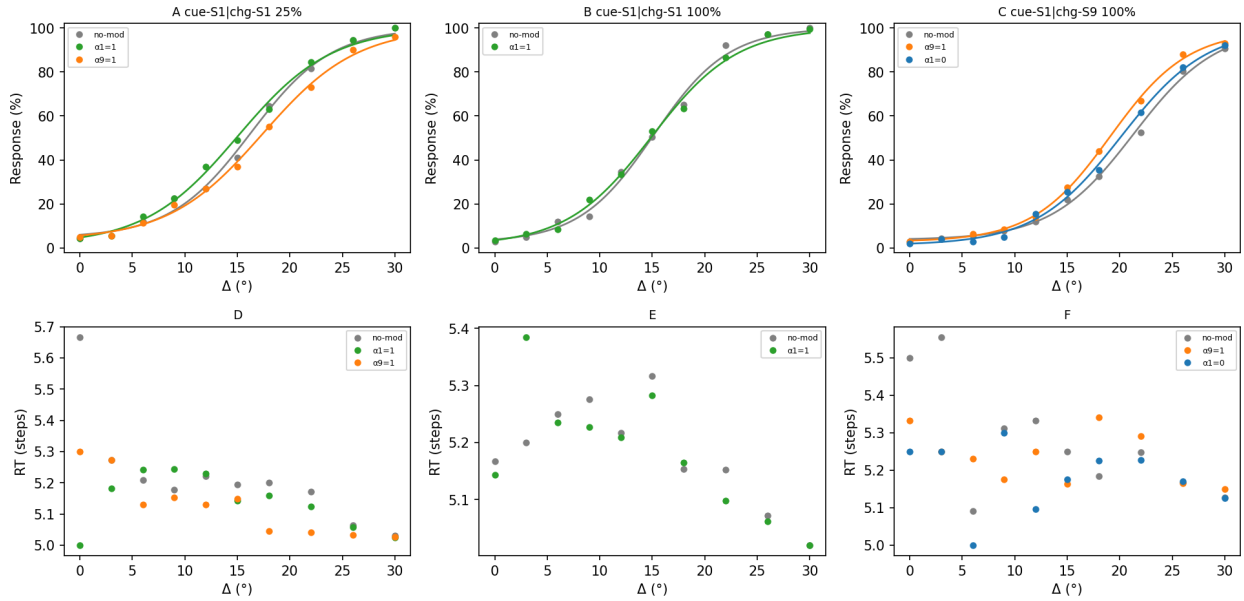


Figure 7: **Attention clamps at t_{change} (9-stim)**. Response (A–C) and RT (D–F) for clamps on S_1/S_9 .

Fig 5 G-K (grid9) — crossattn1: parametric attention manipulation → psychometric params (iter 2999)

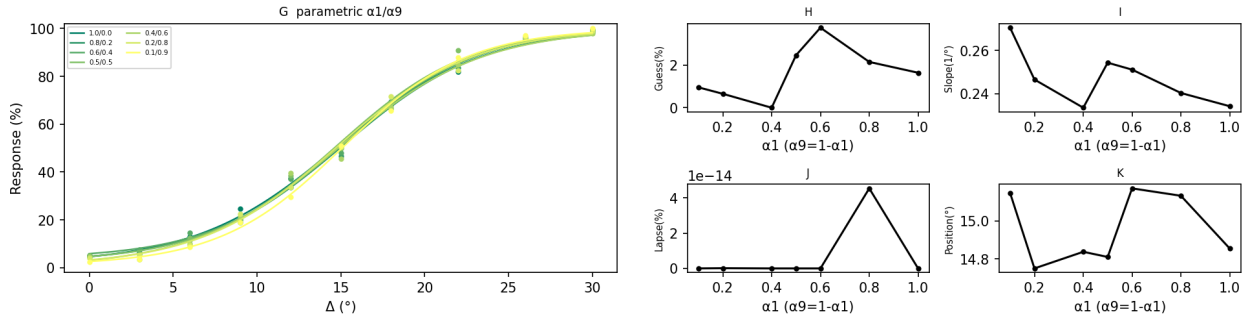


Figure 8: **Parametric manipulation (9-stim)**. Shifting attention $S_1 \leftrightarrow S_9$ grades the psychometric function; fitted parameters vs. α_1 .

Fig 8 --- decode change LOCATION from H , attention on $S1$ (crossattn1, iter 2999; test $n=2000$)

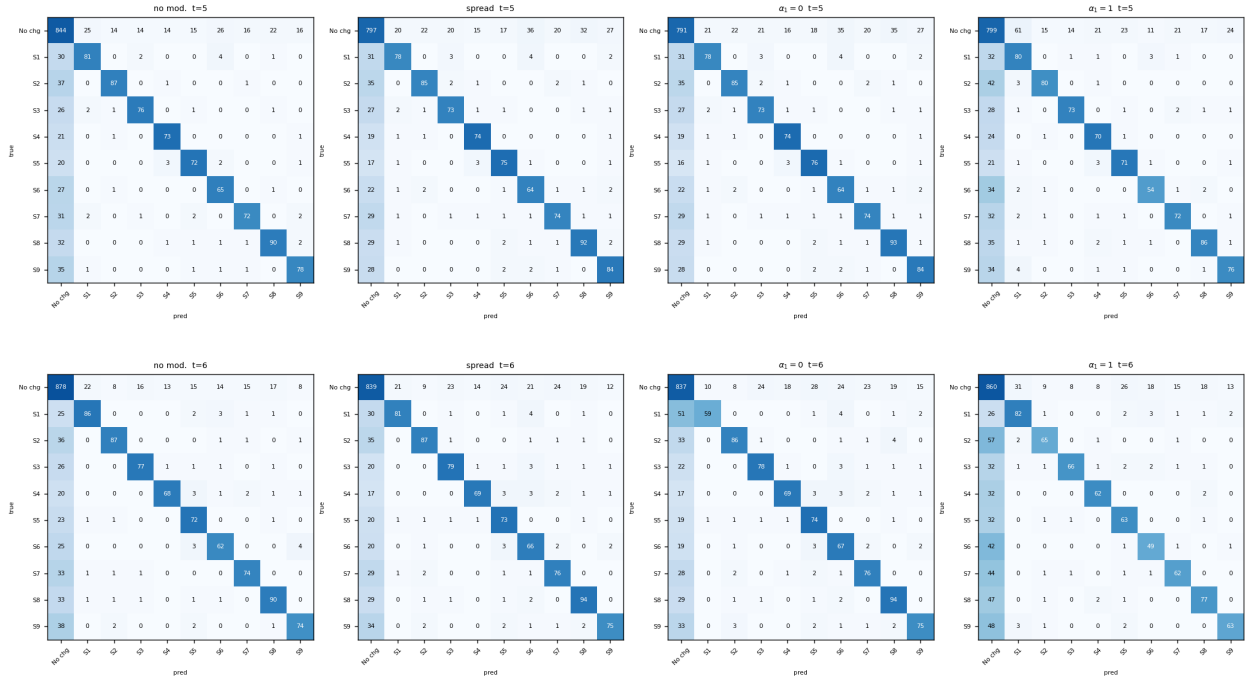


Figure 9: **Fig 8** — decode change LOCATION (10-way) from H , $t=5/6$, under $\{\text{no-mod, spread, } \alpha_1=0, \alpha_1=1\}$.

Fig 9 --- decode change OCCURRENCE from single patch h_1 (crossattn1, iter 2999; $n=2000$)

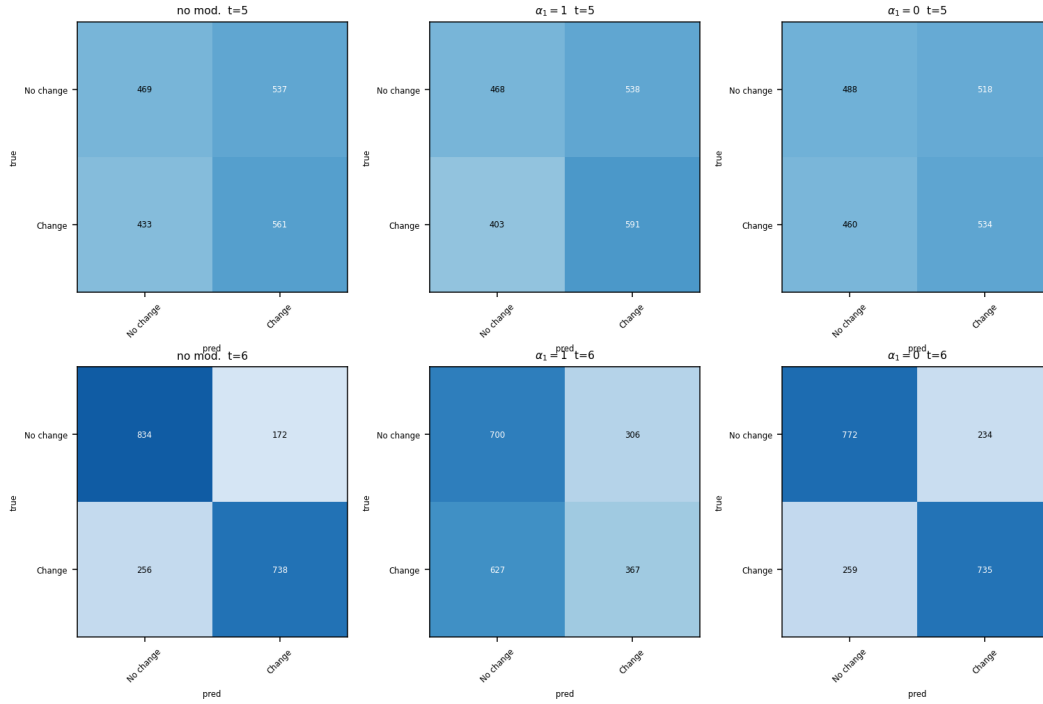


Figure 10: **Fig 9** — decode change OCCURRENCE from single patch h_1 , $t=5/6$.

Fig 10 --- decode from actor first activation layer (crossattn1, iter 2999; n=2000)

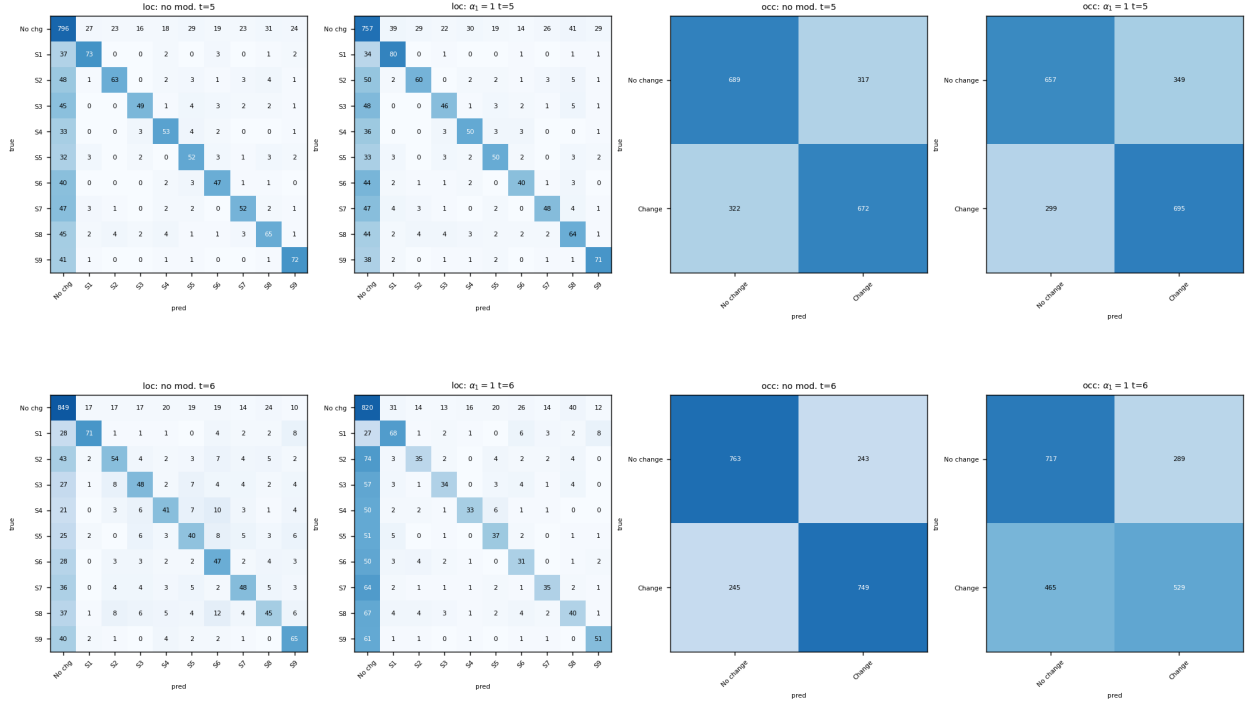


Figure 11: **Fig 10** — decode from actor first activation (location/occurrence).

Fig 11 --- decode S_1 -change vs no-change from actor activation, graded α_1 (crossattn1, iter 2999; n=2000)

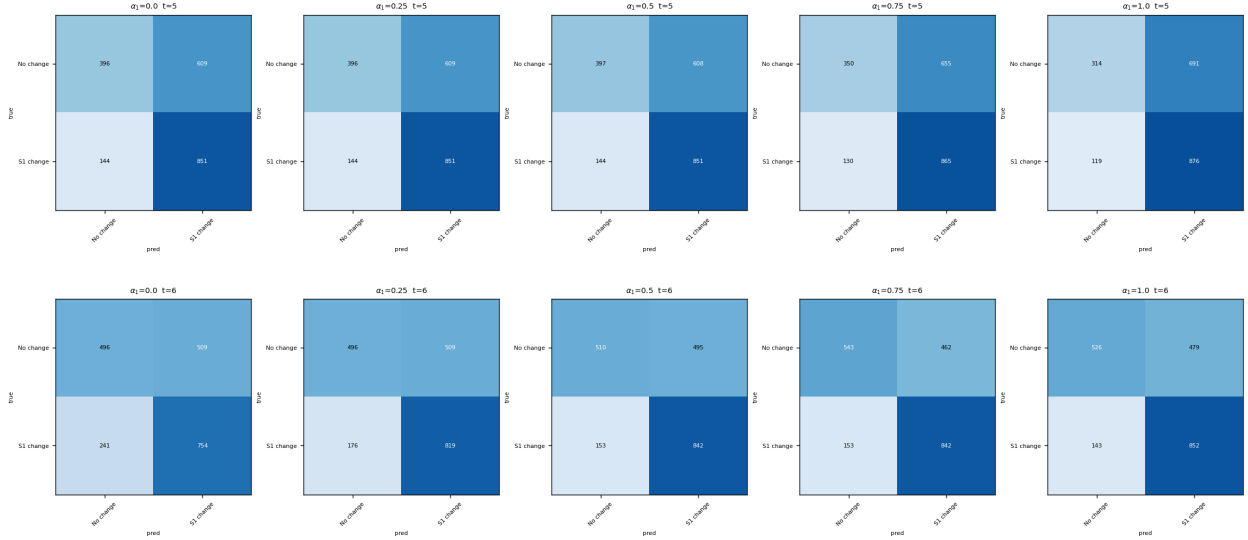


Figure 12: **Fig 11** — decode S_1 -change vs. no-change from actor activation, graded α_1 .

Fig 12 --- actor logit geometry at change frame (crossattn1, iter 2999; n=1500)

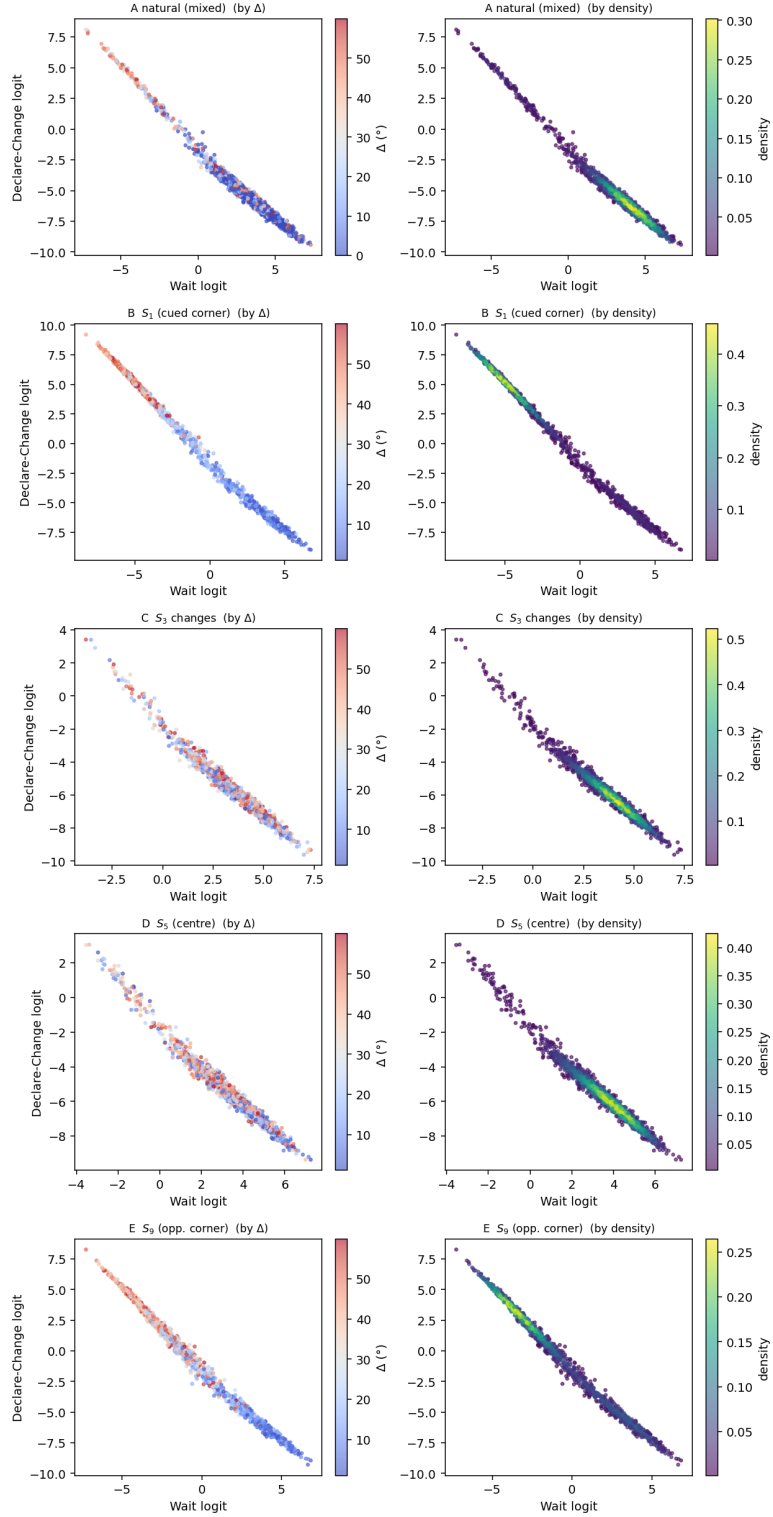


Figure 13: **Fig 12 — actor logit geometry (9-stim).** A: natural. B-E: changes at $S_1/S_3/ S_5/S_9$. Coloured by Δ (left) and density (right).

Fig 13 --- actor logits under S_1 attention modulation (crossattn1, iter 2999; n=1500)

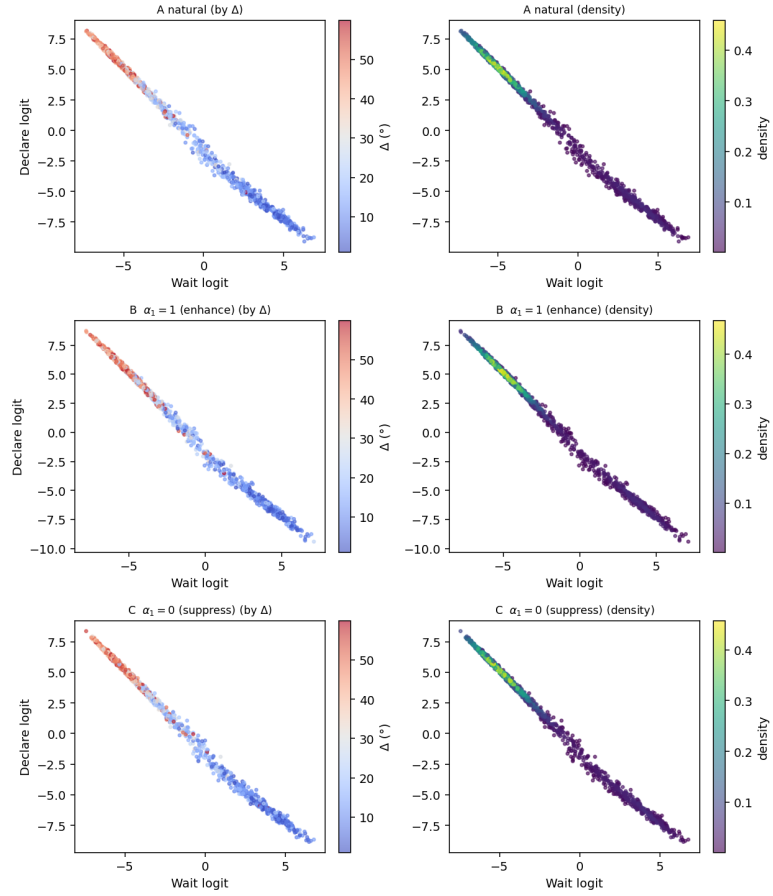


Figure 14: **Fig 13** — actor logits under S_1 attention modulation.

Fig 14 --- TD error & value under change magnitude and attention (crossattn1, iter 2999)

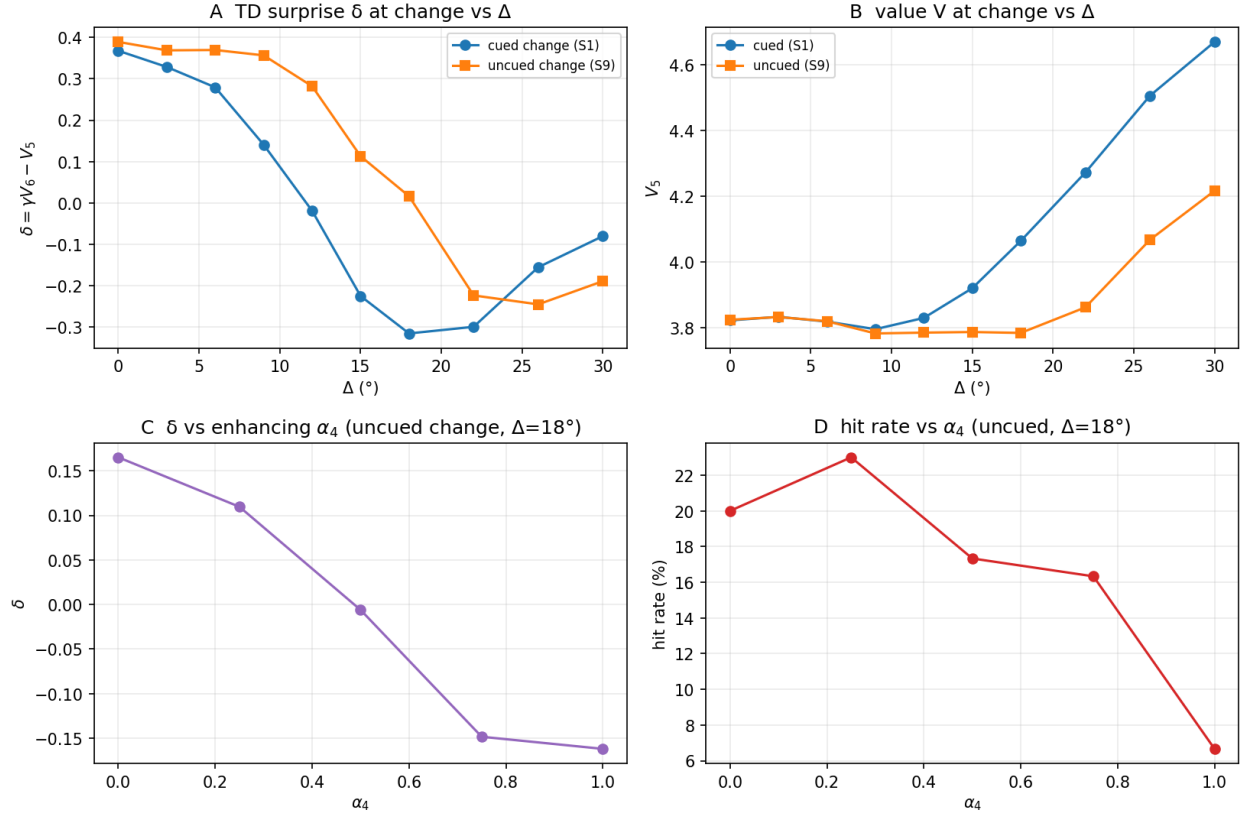


Figure 15: Fig 14 — TD error & value (9-stim).

Fig 15 --- entropy & criterion-value coupling under attention (crossattn1, iter 2999; $\Delta=18^{\circ}$, $n=400$)

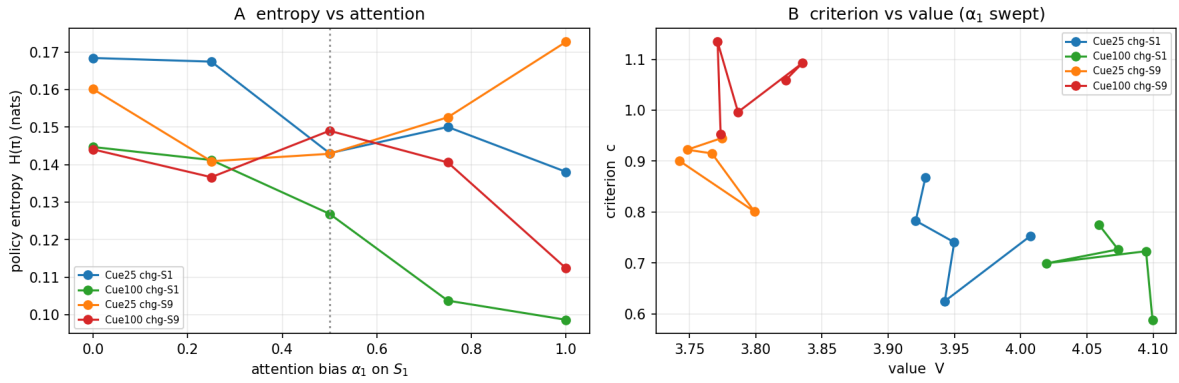


Figure 16: Fig 15 — entropy & criterion-value coupling (9-stim).

Fig 16 --- SDT criterion & sensitivity vs attention bias on S_1 (crossattn1, iter 2999; $\Delta=18^\circ$, $n=300$)

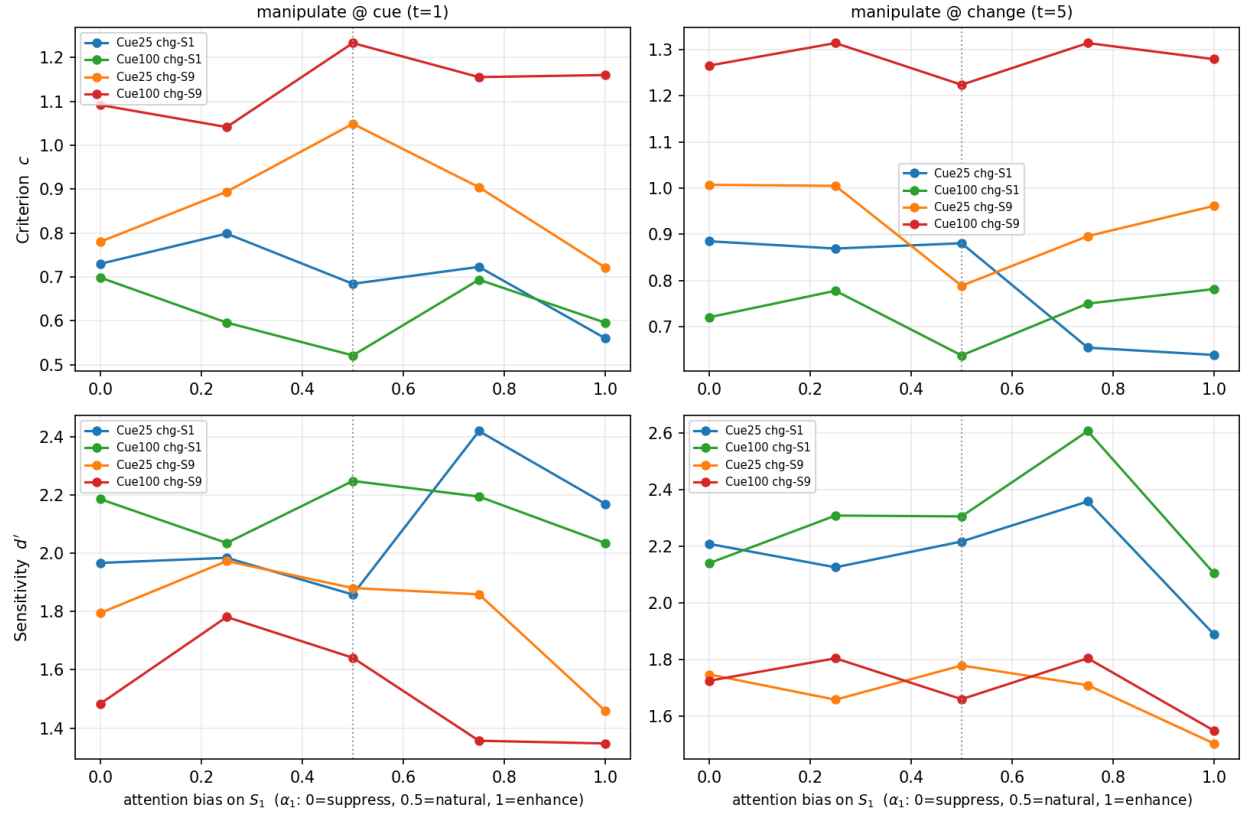


Figure 17: **Fig 16 — SDT criterion & sensitivity vs. attention bias on S_1 (9-stim).** Rows: c , d' . Columns: manipulate at cue-time / change-time. Lines: $\{\text{Cue } 25/100\% \} \times \{\text{change } S_1/S_9 \}$.

Fig 17 analog --- supervised vs RL (RL column reproduced; supervised = out-of-scope baseline) (crossattn1, iter 2999)

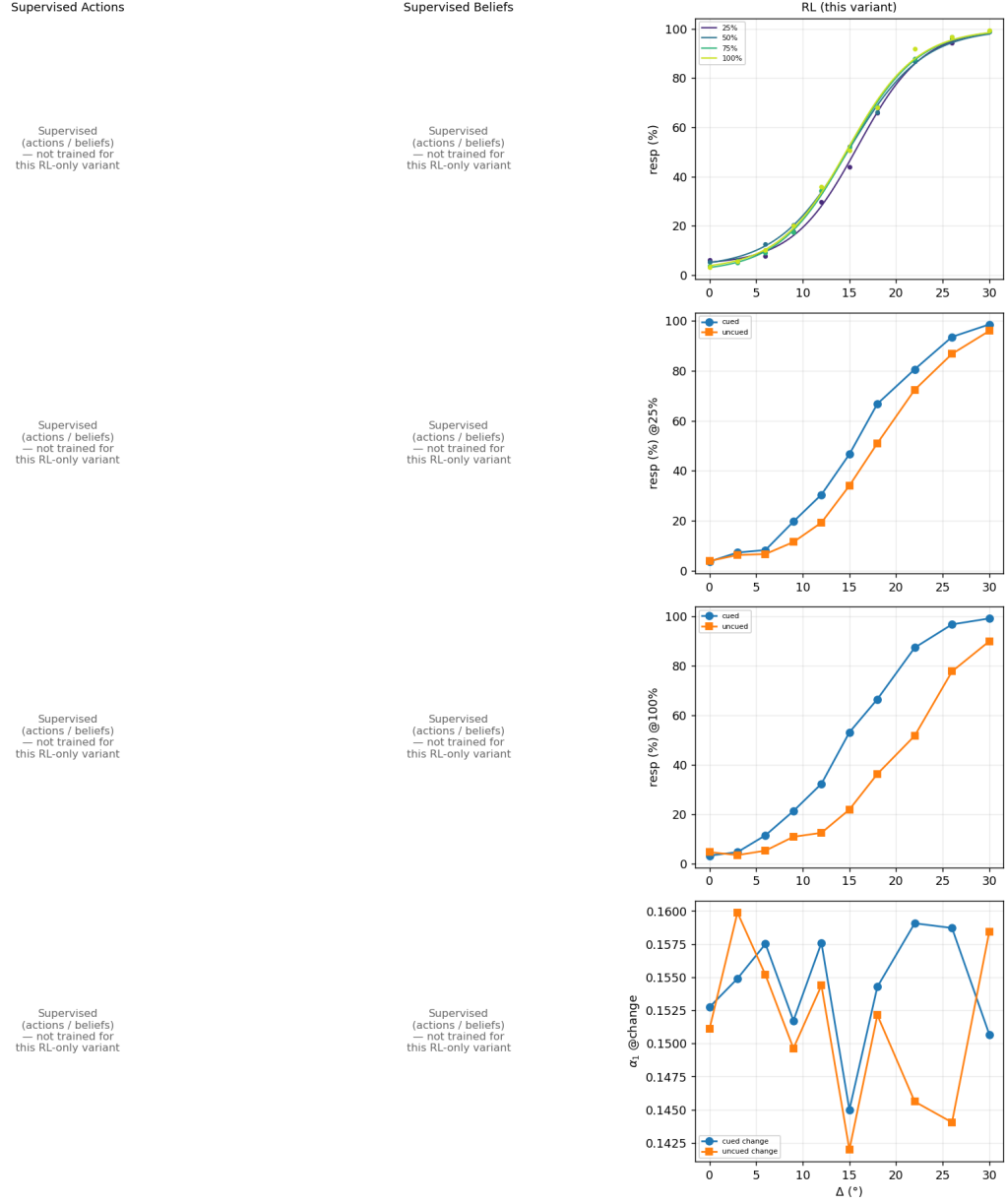


Figure 18: Fig 17 analog — supervised vs. RL (9-stim).